Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, suspension

Species Used Rat, submandibular acini (secretory cells).

Molecules Electroporated Fluorescent dyes (SBFI and MQAE); (3)H-inositol.

Before the Pulse

Cell Growth Medium KRH + Ca

Growth Phase at Harvest Not applicable

Pre-pulse Incubation A variation of the growth medium, but high in potassium and low in sodium and calcium.

Wash Solution Cell growth medium

The Pulse

Instruments Used Not given

Electroporation Temperature Room temperature

Electroporation Medium* Same as pre-pulse

Cell Density 10 mg protein / ml

Volume of Cells 10 mg protein / ml

DNA Concentration 1 mg protein / ml

DNA Resuspension Buffer Same as Pre-Pulse buffer or KRH + Ca

Volume of DNA Not assayed

Cuvette Gap 0.4 cm

Voltage 0.6 to 1.25 kV

Field Strength 1.5 to 3.125 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) Ω none.

Time Constant 0.4 msec

After the Pulse

Outgrowth Medium Not applicable

Outgrowth Temperature Not given

Length of Incubation Not given

Selection Method or Assay Used Not given

Electroporation Efficiency Not given

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

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